Project proposal

Artur Kück

# Self-Imitation Learning for Robotic Cube Stacking

## Objective

Can a robot learn a more reliable final placement by imitating selected movements from its own attempts? I will test whether weighting these movements improves the success rate of a simulated three-cube stacking policy trained by behavior cloning (BC). In a preliminary study, switching between learned controllers for 20 steps during placement changed the final outcome in only 8 of 1,549 starting configurations. This raises a practical question: how often does the robot generate a movement worth imitating?

## State of the art

Oh et al. [1] introduced self-imitation learning to reinforce an agent's own past actions when their return exceeds its value estimate. Critic Regularized Regression [2] uses an action-value estimate to weight the imitation of recorded actions. More recently, Q-Planning [3] used a critic to choose actions for a frozen robot policy at deployment. Here, the critic acts during training: its judgments determine how strongly the policy imitates its own placement attempts.

## Self-imitation learning objective

The policy will collect its own 20-step placement segments using bounded variation in its movement commands. I hypothesize that giving more weight to segments that rescue a failing placement will improve stacking success over uniformly replaying the same data. Replaying from identical simulator states will let me measure how much room there is for improvement and whether a learned critic identifies those opportunities.

## Research plan and evaluation

For each sampled state, I will compare the recorded movement with the frozen BC policy and then let that policy finish both trials. I will count rescued and lost stacks separately. Starting from the same model, I will train three variants on original demonstrations and identical self-generated segments: uniform replay, weighting by the measured outcome difference, and weighting by the critic's estimate. Training budgets and data order will match. I will evaluate all variants and the frozen baseline on the same unseen starts across three training seeds. Success requires the full tower to remain stable for 50 consecutive simulation steps.

## References

[1] Oh et al. (2018). Self-Imitation Learning. ICML.
[2] Wang et al. (2020). Critic Regularized Regression. NeurIPS.
[3] Giridhar et al. (2026). Beyond Imitation: Self-Improving Robot Policies via Off-Policy Q-Planning. arXiv preprint.

Technische Universität Braunschweig

Project schedule

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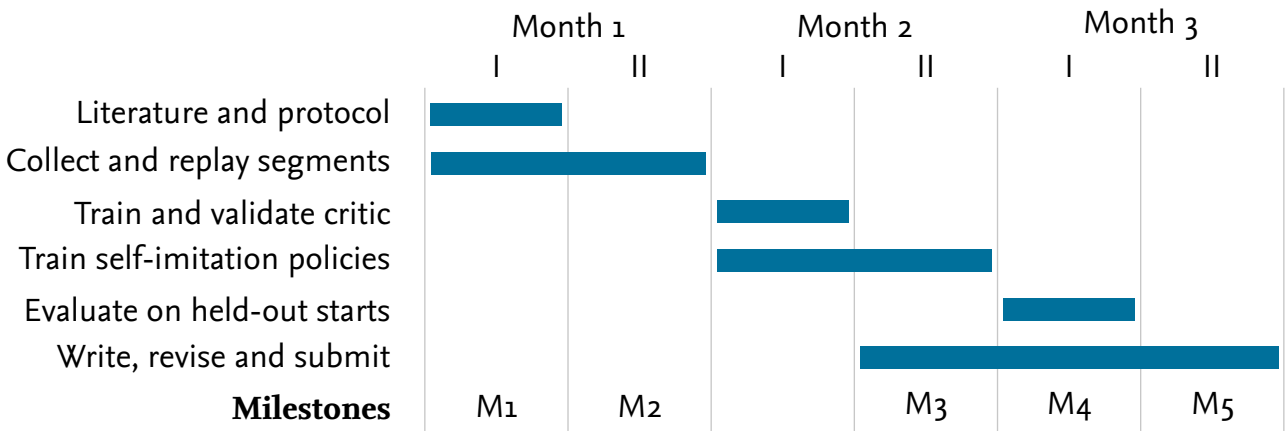
# Work plan and milestones

Three months from the project start. I and II denote the first and second halves of each month.




**M1 Protocol fixed** *Month 1, midpoint*

Specify the exploration bounds, data splits, success criterion and training budget. Verify that replay reproduces the baseline outcome from the same state.

**M2 Dataset frozen** *End of month 1*

Archive the paired segments and report how often the alternative rescues or loses a stack. If rescues are rare, quantify the attainable gain and retain that limitation in the evaluation.

**M3 Models frozen** *End of month 2*

Assess critic rankings on separate start families. Save all three BC variants for each training seed before opening the final test set.

**M4 Evaluation complete** *Month 3, midpoint*

Report paired success-rate differences against uniform replay and the frozen baseline, with results by training seed.

**M5 Project submitted** *End of month 3*

Deliver the report, experiment protocol and scripts needed to reproduce the comparisons.